



Medical Microbiology

Third stage- College of Dentistry

Lab. 16

Mycology

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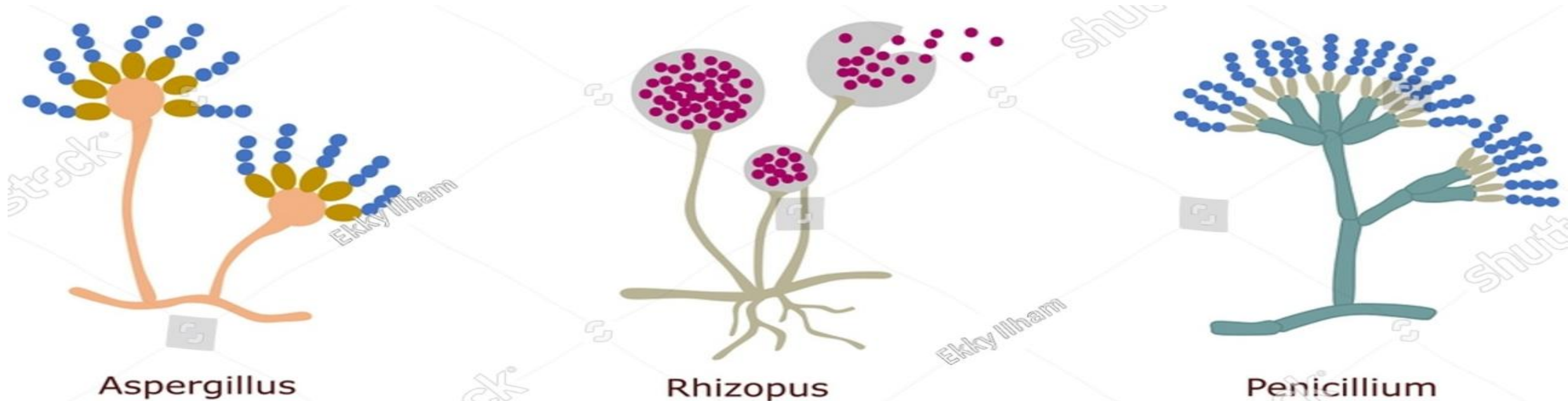
Learning Objectives:

After completing this lab, you should be able to:

1-Describe the microscopic morphology and general Characteristics of Penicillium and Aspergillus

2-Explain the laboratory techniques used for the detection and identification of Penicillium and Aspergillus

3- Describe type of culture Media used for grow the Penicillium and Aspergillus .



Characteristics:

Eukaryotic: Contain a true nucleus, include microscopic organisms such as **yeasts, molds** and **mushrooms**.

Cell Wall: Composed of Chitin (not cellulose).

Nutrition: Fungi are **heterotrophic** organisms. They obtain their nutrition by absorbing organic matter from their environment by secreting enzymes to break down food externally.

Structure: Most fungi consist of long filaments called hyphae, which grow apically.

Non-motile: Most fungi are non-motile.

Reproduction: both sexually and asexually.

Asexual: Occurs through fragmentation of hyphae, budding (common in yeast), or specialized spores like  conidiospores.

Sexual: Involves three main stages:

- Plasmogamy.
- Karyogamy.
- Meiosis to produce genetically varied spores.



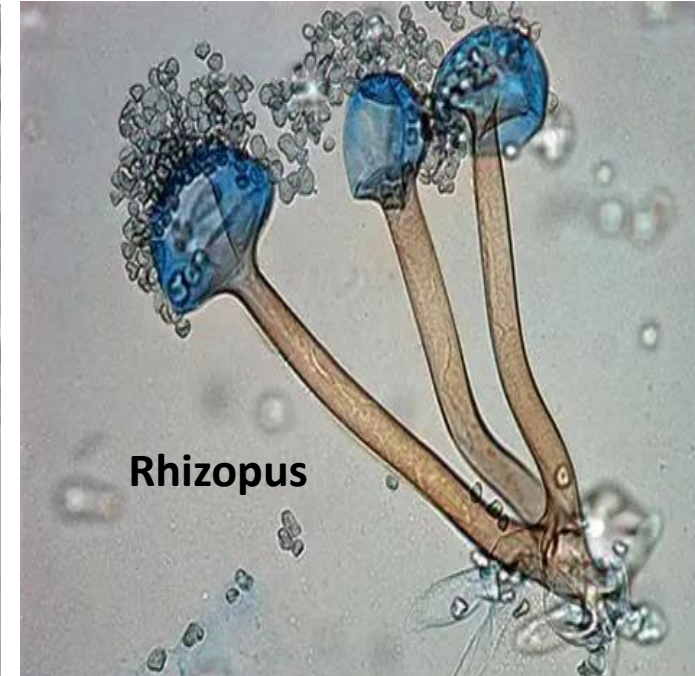
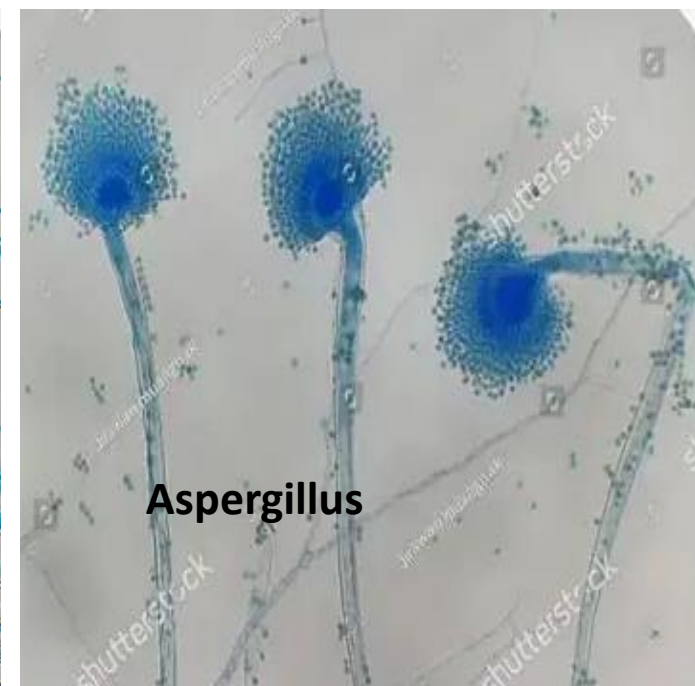
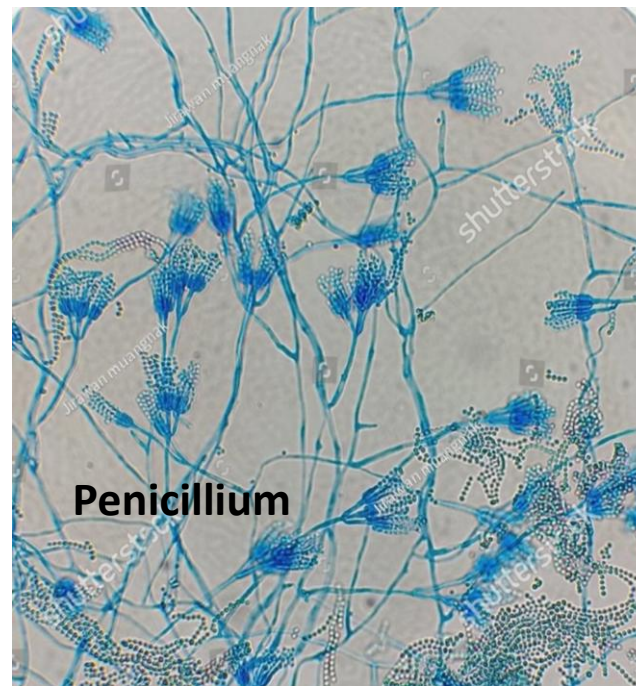
Common Genera:

Aspergillus: Characterized by a "Vesicle" (mop-like appearance).

Penicillium: Characterized by a brush-like appearance (Broom-like).

Candida albicans: Shows budding yeast cells and Pseudo hyphae (false hyphae).

Rhizopus: Known as bread mold; features root-like structures called Rhizopus.



Penicillium:

- **found in:** soil, air, and food.
- **Positive Impacts on Humans:**

1-Antibiotic Production: It is the primary source of Penicillin, the first antibiotic discovered by Alexander Fleming.

2-Food Industry: (e.g., *P. roqueforti*) are used in the production of cheeses, providing unique flavors and textures.

-Negative Impacts

Allergies: Caused by inhaling airborne spores, especially in damp environments.

Symptoms include:

Respiratory: Runny or stuffy nose, sneezing, and itchy throat.

Ocular: Red, itchy, or watery eyes (conjunctivitis).

Dermatological: Skin rashes upon contact.

Asthmatic: Shortness of breath, and a persistent dry cough.



Opportunistic Infections: certain species (*Penicillium*) can cause severe systemic infections in immunocompromised patients. **Fever and Chills, Skin Lesions, Weight Loss, Organ Enlargement, Respiratory Distress.**

Mycotoxins: Some species produce toxic secondary metabolites, such as **Patulin** or **Citrinin**. If these contaminate food, they can lead to kidney or liver damage.

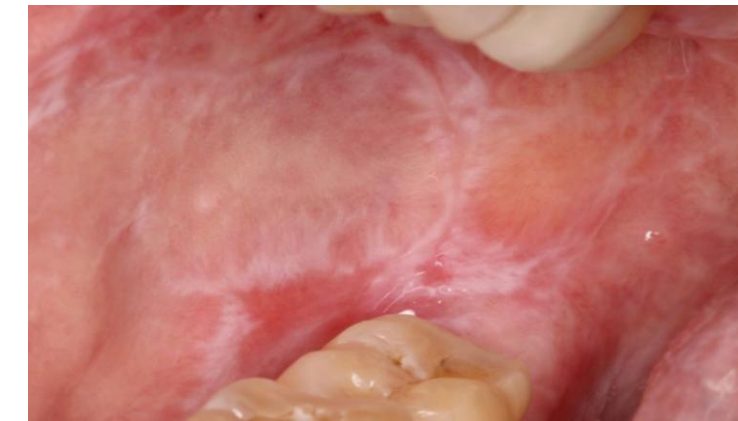
Oral and Gingival Symptoms:

cause in very rare cases or in people with severely weakened immune systems, symptoms similar to other oral fungal infections (like *Candida*) may appear, including:

White or Creamy Patches: appearing on the tongue, inner cheeks, or gums.

Redness and Swelling: The affected tissues may look red, inflamed, and might bleed slightly when touched or brushed.

Pain and Difficulty Eating: A burning sensation in the mouth, making swallowing or eating solid foods painful.



Laboratory Diagnosis Methods:

☐ Culture Media

- To cultivate *Penicillium* in the lab, we use **Carbohydrate-rich media** with a slightly acidic pH.
- **PDA (Potato Dextrose Agar):** The most common medium; ideal for observing colony morphology.
- **Czapek Dox Agar:** A semi-synthetic medium used for detailed morphological studies and species identification.
- **MEA (Malt Extract Agar):** Used to enhance and stimulate sporulation (spore production).

PDA



Czapek Dox Agar



MEA



Colony Morphology: after incubation for 3-5 days at 25°C.

Color: Usually shades of green, blue-green, or greyish-green.

Texture: Velvety, woolly, or granular.

Margin: surrounded by a thin **white border**.

Microscopic Examination:

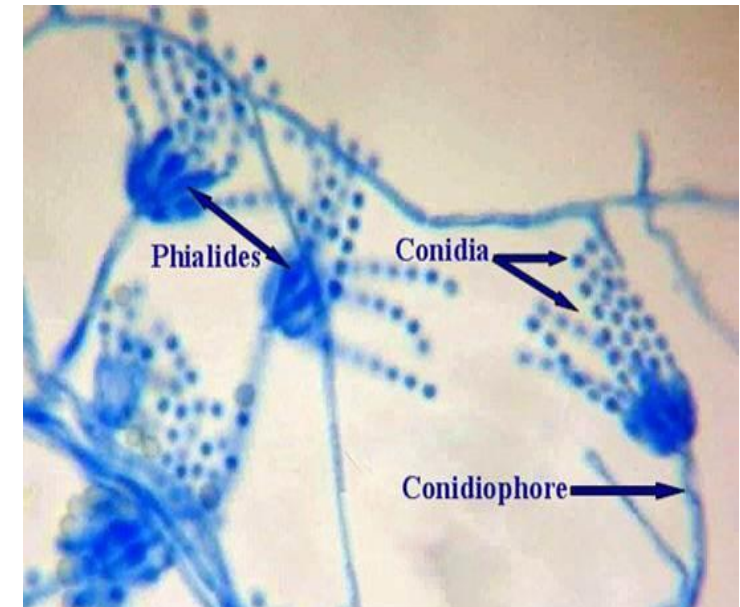
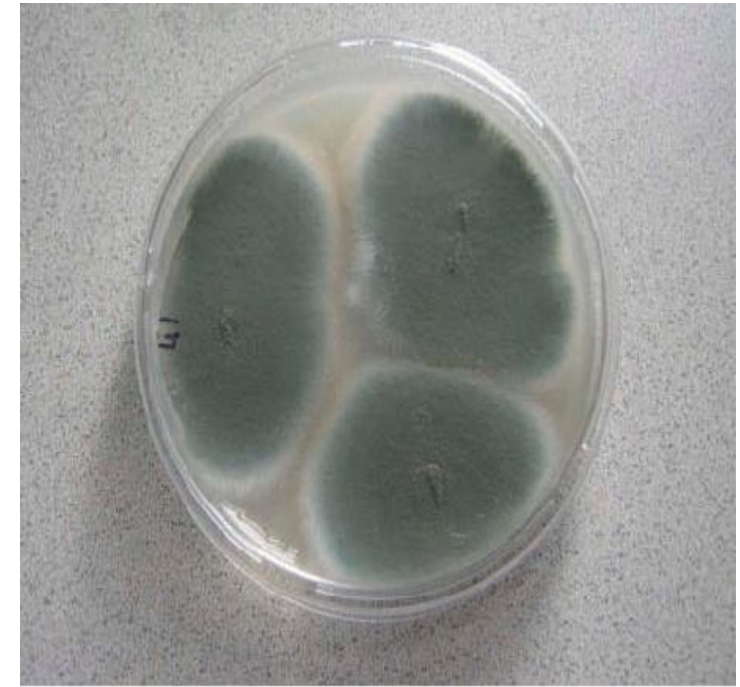
We use **Lactophenol Cotton Blue (LPCB) stain** to visualize the structures. Under the microscope, Penicillium has a characteristic "**Brush-like**" appearance consisting of:

Conidiophores: Branched or unbranched stalks.

Metulae: Secondary branches arising from the conidiophore.

Phialides: Flask-shaped structures that produce the spores.

Conidia: Round spores arranged in long chains.



Aspergillus:

A type of mold (fungus) found everywhere in the environment.

Its effects on the mouth and gums:

1-Painful ulcers: Covered with grey or black dead tissue.

2-Dark patches: Black or discolored areas on the palate or gums (tissue death).

3-Swollen gums: Severe inflammation with pain.

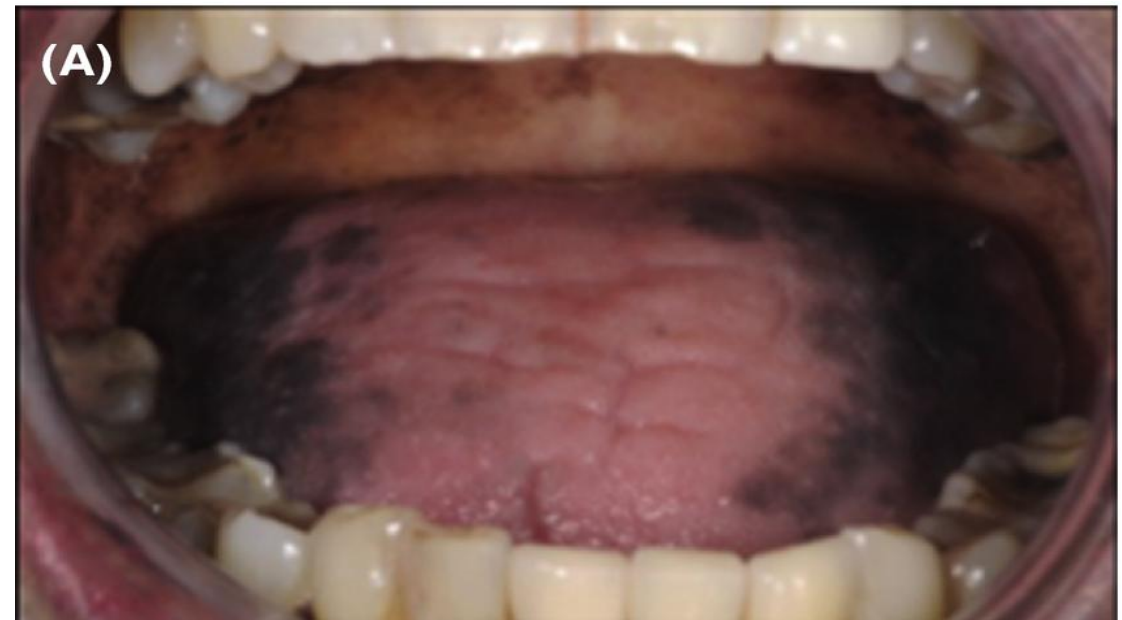
4-Loose teeth: damage to supporting tissues.

5-Jaw infection: May spread to bone causing deep pain (osteomyelitis).

6-Upper tooth pain: Sometimes comes from sinus infection, not the teeth directly.



Necrotic Ulcerative Lesions



Tissue Discoloration



Maxillary Osteomyelitis



Acute Gingival Swelling

Laboratory Diagnosis Methods:

☐ Culture Media:

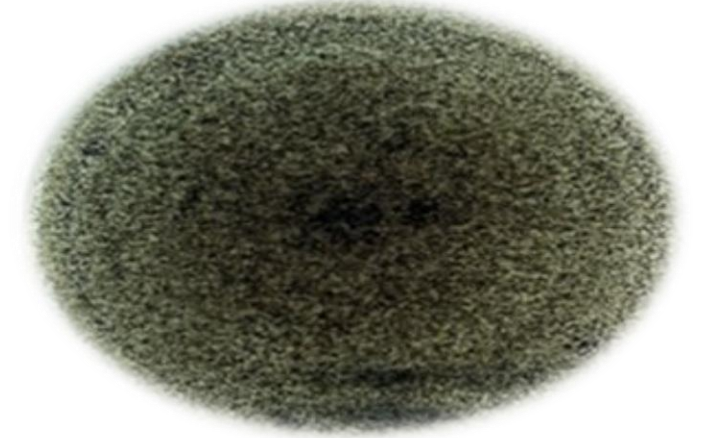
Usually incubated at 25°C to 37°C,
Colonies typically become visible within
3 to 5 days.

- **SDA (Sabouraud Dextrose Agar):** The medium for primary isolation due to its acidic pH, which inhibits bacterial growth.
- **PDA (Potato Dextrose Agar):** Used to promote rapid sporulation (spore formation).
- **Czapek Dox Agar:** it helps define colony color and growth patterns accurately.

SDA



PDA



Czapek Dox Agar



Colony Morphology:

Color: Distinctive for each species Smoky-green ,Coal-black, Yellow-green.

Texture: Usually Granular, velvety, or powdery due to dense spore production.

Microscopic Examination:

Using the **Lactophenol Cotton Blue (LPCB) staining** technique, the fungus shows a **radiating head** structure **consisting of:**

Conidiophore: A long, unbranched stalk.

Vesicle: A swollen, bulbous tip at the end of the conidiophore.

Phialides: Flask-shaped cells arising from the vesicle.

Conidia: Chains of round spores covering the vesicle partially or entirely.





جامعة الكفيل
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Any Question?



Thank you